Mids Syllabus

Cormen:
Chapter 1
(Introduction)
Chapter 2
(Insertion sort and merge sort [Selection sort given as assignment] detailed analysis and proof of correctness using loop invariant)
Chapter 3 (3.1 and 3.2 [Big O, Big Omega, Big theeta only])
(Asymptotic Notations overview and usage, solving asymptotic notations for c and n)
Chapter 4 (4.3, 4.4, 4.5)
[Solving Recurrences (substitution (induction), recurrence tree, master theorem)]

Anany:
Chapter 1 (1.1, 1.2, 1.3 & 1.4 optional)
[Introduction]
Chapter 2 (2.1, 2,2, 2.3, 2.4)
[Analysis Frame work and analysis of recursive and iterative algorithms using basic operation]
Chapter 3 (3.1)
[Selection and Bubble sort] **for ref on basic op*
Chapter 4 (4.1)
[Insertion Sort] **for ref on basic op*
Chapter 5 (5.1)

[Meger Sort] **for ref on basic op*

Contents Studied in lectures

Week 1 and 2:

Algorithms- an Overview:

- Introduction to algorithms
- Role of Algorithms in Computing
- Fundamentals of Algorithmic Problem Solving

Fundamentals of Analysis of Algorithms Efficiency

- Analysis Framework
- Measuring an Input SIze
- Unit of Measuring Runtime

Cormen:
Chapter no 1 (1.1 and 1.2)
 Assignment Questions:1.2-2, 1.2-3, 1-1
A. Leviton:
Chapter no 1 (1.1 and 1.2) Reading 1.3 and 1.4 (for your interest)
Chapter no 2 (2.1 till pg 45)

Week # 3

Decrease and Conquer technique with a simple example algorithm:
Insertion Sort Pseudocode and dry run

Proof of Correctness: Loop Invariant
Analyzing algorithms with RAM Model (Best, Average, Worst)
Cormen chapter no 2: 2.1 & 2.2
A. Leviton chapter no 4: 4.1
Divide and Conquer technique with a simple example algorithm:
Merge Sort Pseudocode and dry run
Analyzing algorithms with RAM Model (Best, Average, Worst)
Cormen chapter no 2: 2.3
A. Leviton chapter no 5: 5.1
Practice Questions:
Cormen 2.1-1, 2.1-3, 2.2-2, 2.3-1, 2.3-2

Week # 4

Asymptotic Notations:

Big O, Big Omega, Big Theeta
Cormen: Chapter 3 (3.1, 3.2 defn only)
Anany: Chapter 2 (2.2 till page 55)
Practice Questions:
Cormen:2.2-1, 3.2-2, 3.2-3
Anany: 1,2,3,5

Week # 5, 6

Mathematical Analysis of non-recursive algorithms
Basic Operation Identification
Expressing the loops in summation
Anany chapter no 2: 2.3

Mathematical Analysis of Recursive algorithms
Basic Operation Identification
Expressing the recurrence in terms of basic operation
Anany chapter no 2: 2.4

Solving Recurrences
1. Substitution Method
2. Recurrence Tree method
3. Master Theorem
Cormen chapter no 4: 4.3, 4.4 and 4.5